Harder Geometrical Problems — problem 12.3.2.99

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Let $ABCDEF$ be a convex hexagon such that $AB \parallel DB$, $BC \parallel EF$ and $CD \parallel AF$. Let R_A, R_C, R_E denote the circumradii of triangles FAB, BCD, DEF, respectively, and let P denote the perimeter of the hexagon. Prove that

$$R_A + R_C + R_E \geq \frac{P}{2} \qquad \text{(IMO 1996)}.$$